

PreventNet System Overview

A preventive-medicine review where a GP practice, a pharmacy and a lab each keep their patient records, answer only short purpose-tagged questions, and train a diabetes-risk model together, all without pooling data. A coordinator combines the answers, code verifies them, and a clinician approves before anything is released.

3 departments, each with its own data	10 allowed questions (1 always refused)	6 model calls per SuperGrid run (cap 8)	768 people in the public Pima dataset	0.79 federated model test AUC
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1 The problem

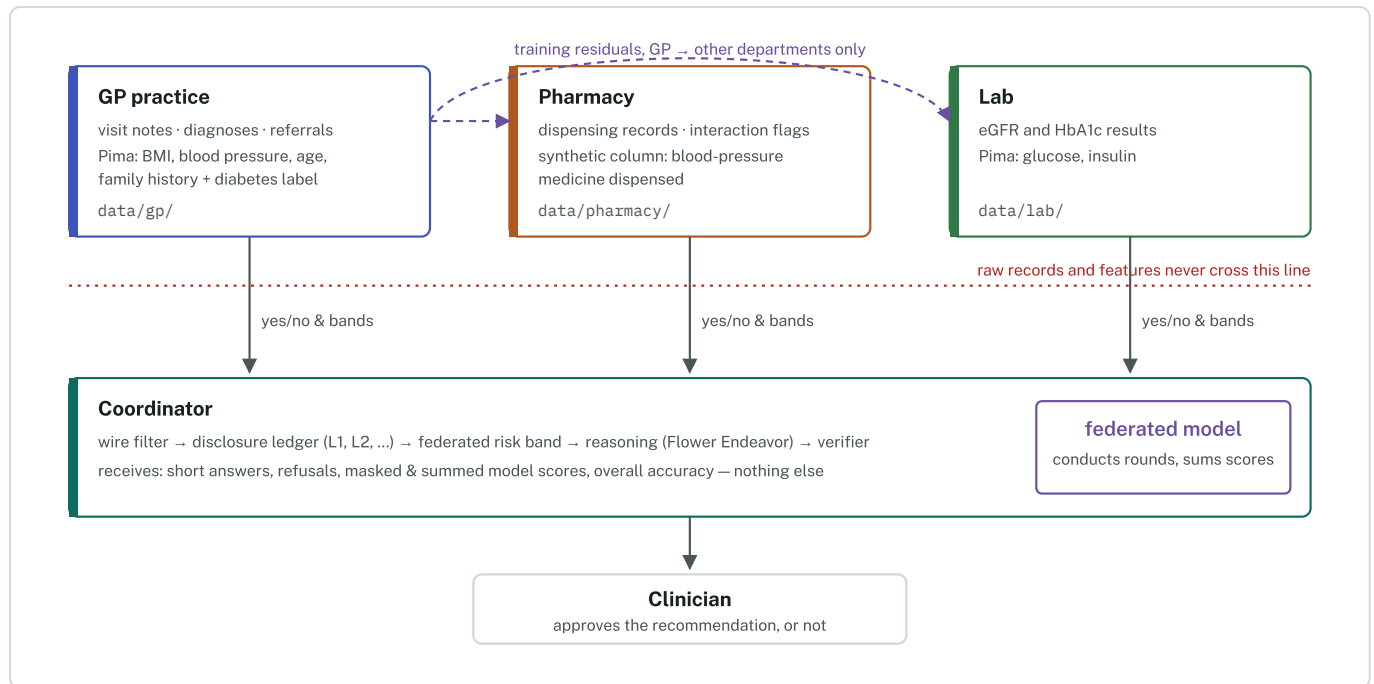
By law, each institution may use a patient's data only for its own purpose. That protects patients, but it also hides risks that only appear when you combine what several institutions know. PreventNet's example patient is **P001**:

HOLDER	WHAT IT KNOWS ABOUT P001	ALONE, THIS LOOKS...
GP	High blood pressure since 2019. Mentioned taking ibuprofen for knee pain. Contrast CT scan booked for 5 July.	Routine
PHARMACY	Metformin dispensed monthly. Ibuprofen 400 mg dispensed on 4 June. Its own interaction checker flagged the pair.	Routine
LAB	Kidney function (eGFR) fell from <i>normal</i> (2024) to <i>moderately reduced</i> (2026). HbA1c 5.9%, in the <i>prediabetic</i> band.	Worth watching

Together, the picture is weakening kidneys, plus metformin, plus an anti-inflammatory painkiller, plus contrast dye in three weeks. That is a well-known, preventable route to acute kidney injury. The right action is a medication and kidney-safety review *before* the scan. PreventNet's job is to surface that for a clinician without anyone handing over their records.

2 How the system is built

There is one application, the **PreventNet AgentApp**, which runs on Flower's cloud (SuperGrid). A set of **four local deployments** on your laptop runs the same logic as separate programs, so you can watch each part at work.



THE PRODUCT

SuperGrid AgentApp

Started with `flwr run . supergrid`. Runs in Flower's cloud with the Flower Endeavor model (`flower-endeavor-v1.0`).

- Each department agent is a separate model conversation that sees only its own records.
- The three departments train the diabetes model inside the run.
- It prints one `PREVENTNET_RESULT` line and saves the result in the run's state.

THE WINDOW INTO IT

Four local deployments

Started with `uv run python -m local.launch`. GP, Pharmacy and Lab each run as their own server with their own page; the fourth is the coordinator dashboard.

- Departments answer with transparent rules, so no AI reads raw records locally.
- The dashboard can launch the SuperGrid app and stream its output live.

3 The review workflow, step by step

This is the order of events in one SuperGrid run for one patient. Each step lists who acts and exactly what leaves that step.

01 **COORDINATOR** Plan the questions

The model picks relevant questions from the fixed list of 10. The full list is always used, so the demo is repeatable. `model call 1`

Leaves: **question IDs only**. No patient data is involved yet.

02 **GP** **PHARMACY** **LAB** Policy check first

Each department checks every question's purpose against its own disclosure policy *before* looking at any record. The marketing question is refused here on every run.

Leaves: **a refusal** (question, department, reason) for anything out of purpose.

03 **GP** **PHARMACY** **LAB** Answer from own records

Each department agent is a separate model conversation that sees only its own `records.jsonl` and answers in strict JSON. `model calls 2-4`

Leaves: **nothing yet**. The answers still have to pass step 4.

04 **GP** **PHARMACY** **LAB** Wire filter

`strip_for_wire()` allows only field, value, purpose, source and time. It rejects notes, names, dates of birth, record IDs, unknown keys, and any text longer than 40 characters.

Leaves: **short facts** such as `egfr_band = moderately_reduced`.

05 **COORDINATOR** Write the disclosure ledger

Every fact gets an ID (L1, L2, ...) and every disclosure is logged as a run event. The ledger is the coordinator's whole view of the patient.

Holds: **about 9 facts + 1 refusal** for P001.

06 **FEDERATED** Train the diabetes model together

The GP, Lab and Pharmacy each compute their part of a logistic-regression score from their own Pima columns, over 60 rounds (details in section 7). The patient's risk is then scored the same way. No model calls.

Leaves: **masked, summed scores and overall accuracy**, then one ledger fact: `fl_diabetes_risk_band`.

07 **COORDINATOR** Reason over the ledger

Flower Endeavor reads *only* the ledger and writes RISKS (each citing ledger IDs), one preventive RECOMMENDATION, and a CONFIDENCE level. It is instructed never to diagnose or change treatment.

`model call 5, streamed`

Leaves: **the risk summary text**.

08 **VERIFIER** Check with code

Code strikes any risk line without a valid citation, applies two contradiction rules, and re-checks every fact against the wire filter. It then sets the verdict. The model only writes a two-sentence explanation.

`model call 6`

Verdict: **APPROVED_FOR_REVIEW** **ESCALATE_CONFLICT** **BLOCKED_UNVERIFIED**

09

CLINICIAN Approve, or resolve upstream

The recommendation stays blurred until a clinician clicks **Approve & send to GP**. On a conflict, approval is disabled: the institutions must reconcile their records first.

Released: **one approved preventive action**, recorded with name and time.

4 What the coordinator sees

The screenshot shows the full result for P001: 10 facts (L1–L10), one refusal, and nothing else. The department cards show only how many records each department scanned. The records themselves stay local.

Patient P001

local deterministic rules (offline)

APPROVED FOR REVIEW

engine: local deployments + offline rules

federated diabetes risk (local deployments, test AUC 0.79): **21% · moderate**

GP

6

raw records: LOCAL ONLY

PHARMACY

8

raw records: LOCAL ONLY

LAB

6

raw records: LOCAL ONLY

Everything the coordinator ever saw: 10 items — all listed below.

id	role	field	value	purpose
L1	gp	chronic_conditions	hypertension	renal-risk-review
L2	gp	contrast_imaging_planned_90d	true	renal-risk-review
L3	gp	patient_reported_otc_nsaid	true	medication-safety-review
L4	pharmacy	nsaid_dispensed_90d	true	medication-safety-review
L5	pharmacy	metformin_active	true	renal-risk-review
L6	pharmacy	interacting_disperse_flag	true	medication-safety-review
L7	lab	egfr_band	moderately_reduced	renal-risk-review
L8	lab	egfr_trend_24m	declining	renal-risk-review
L9	lab	hba1c_band	prediabetic	metabolic-screening
L10	federated-model	fl_diabetes_risk_band	moderate	metabolic-screening

⊘ refused q_gp_marketing_segment for gp — out_of_purpose

Risk summary

Review result. Every row is a short fact with its source and purpose. L10 comes from the federated model. The red row is the marketing question, refused by policy.

RISKS

- Reduced kidney function (eGFR moderately_reduced, trend declining) ahead of planned contrast imaging [L7] [L8] [L2]
- Active metformin with planned contrast imaging [L5] [L2]
- NSAID exposure alongside reduced kidney function [L3] [L4] [L7]
- Pharmacy interaction check flagged a dispensed drug pair [L6]
- Metabolic risk: HbA1c band prediabetic, federated diabetes-risk model moderate [L9] [L10]

RECOMMENDATION

Invite P001 for a pre-imaging medication and renal-safety review (eGFR re-check, NSAID and metformin reconciliation) before the scheduled contrast study.

CONFIDENCE

high

Risk summary. Every risk cites the ledger IDs it rests on, such as [L7] [L8] [L2] . A line without a valid citation would be struck through by the verifier.

5 The data

Each department folder holds two kinds of data:

- **Review records** (`records.jsonl`): synthetic visit notes, dispensing records and lab results for three patients under review (P001–P003) plus background patients.
- **Cohort columns** (`cohort.jsonl`): that department's share of the public **Pima Indians Diabetes** dataset.

The Pima data comes from the US National Institute of Diabetes and Digestive and Kidney Diseases. It covers 768 women aged 21 and over of Pima Indian heritage, with no names or IDs, and records whether each developed diabetes within five years. It is split into 615 people for training and 153 held out for testing. Several columns record “not measured” as 0; those are counted below as missing and filled in by each department on its own.

HOLDER	COLUMN	MEANING	MEAN	RANGE	MISSING
GP	<code>pregnancies</code>	Number of times pregnant	3.85	0–17	0
	<code>blood_pressure</code>	Diastolic blood pressure (mm Hg)	72.4	24–122	35
	<code>skin_thickness</code>	Triceps skinfold (mm)	29.2	7–99	227
	<code>bmi</code>	Body mass index (kg/m ²)	32.5	18.2–67.1	11
	<code>diabetes_pedigree</code>	Family-history score	0.47	0.08–2.42	0
	<code>age</code>	Age (years)	33.2	21–81	0
	<code>outcome</code>	Label: developed diabetes within 5 years. Held only by the GP.			268 of 768 (34.9%)
LAB	<code>glucose</code>	Plasma glucose, 2 h oral glucose test (mg/dL)	121.7	44–199	5
	<code>insulin</code>	2-hour serum insulin (μU/mL)	155.6	14–846	374
PHARMACY	<code>antihypertensive_dispensed</code>	Synthetic. Blood-pressure medicine dispensed (Pima has no medication data)	0.19	0–1	0

Means and ranges exclude the missing values. These figures are aggregates that each department computes itself; the dashboard shows the same table.

6 The questions and where answers come from

None of these fields exist in the Pima dataset. Each department derives the answer from its own review records and sends back only the short value. Pima's only job is to train the federated model, whose output enters the ledger as one extra fact (last row).

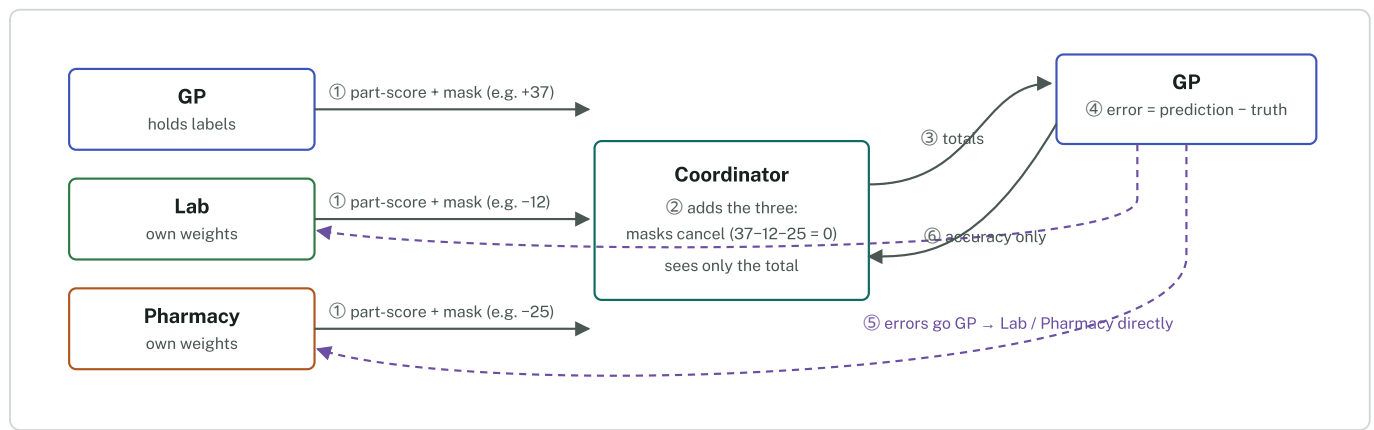
FIELD	ASKED OF	DERIVED FROM	WHY IT MATTERS
chronic_conditions	GP	Diagnosis records	High blood pressure and diabetes are the main causes of kidney disease.
contrast_imaging_planned_90d	GP	Referral records mentioning contrast	Contrast dye can injure weak kidneys; metformin is often paused around it.
patient_reported_otc_nsaids	GP	Visit notes mentioning ibuprofen, naproxen or NSAIDs. The note itself never leaves.	Shop-bought painkillers may not appear in pharmacy records, and they reduce kidney blood flow.
nsaid_dispensed_90d	PHARMACY	NSAID dispensed within 90 days of the review date	An independent second source. Disagreement with the GP is a data-integrity signal.
metformin_active	PHARMACY	Metformin dispensed within 90 days	Metformin, weak kidneys and contrast together call for a pre-imaging review.
interacting_dispense_flag	PHARMACY	The pharmacy's own interaction-check record	Its safety system already saw a risky drug pair.
egfr_band	LAB	Latest eGFR result, stored as a clinical band	The core kidney weakness behind this harm.
egfr_trend_24m	LAB	First vs. latest eGFR band over 24 months	Kidneys getting weaker make a new insult more dangerous.
hba1c_band	LAB	Latest HbA1c result, stored as a band	Prediabetes is the window where prevention works.
insurance_marketing_segment	GP	Nothing. Policy refuses it before any record is read.	Proves that out-of-purpose questions are refused on every run.
fl_diabetes_risk_band	FEDERATED	Federated logistic regression over the Pima columns	A population-trained diabetes risk that no single department could compute alone.

7 Federated learning

The goal is to predict diabetes risk from columns that sit in three different departments, without any department revealing its columns. The model is **logistic regression**, because its score is a plain sum:

```
risk score = GP part + Lab part + Pharmacy part      probability = 1 / (1 + e^(-score))
              (BMI, BP, age...)  (glucose, insulin)  (dispensing flag)
```

Each department computes its own part from its own columns and keeps its own weights. Only the parts are combined, which is what lets the model split cleanly across departments that hold different columns. The weights are also easy to read: each department's page shows its own.



- **Masks.** The departments share a secret the coordinator never gets. Each adds a random mask to its part-scores, and the three masks cancel when summed, so the coordinator can't isolate any one department's contribution. Each mask is used once only.
- **Errors bypass the coordinator.** The GP sends them straight to the Lab and Pharmacy, and every department updates its own weights (step ⑦, not drawn).
- **Alignment check.** Before training, the departments compare fingerprints of their patient-ID lists without revealing the lists.
- **Where it runs.** In the SuperGrid run, the three departments are separate objects inside one task. Locally, they are three separate servers talking over HTTP. The code is the same.

Federated learning · diabetes risk

Logistic regression trained across the departments on the Pima data: each department holds different columns for the same people. The coordinator starts each round and adds up the masked scores, but all training happens inside the departments.

Rounds Learning rate Train across the 3 local deployments

Showing 60 rounds (local deployments). Test AUC 0.788.

Trained across the three local deployments over HTTP at 2026-09-16T12:46:52Z.

0.788

Test AUC

74.5%

Test accuracy

0.452

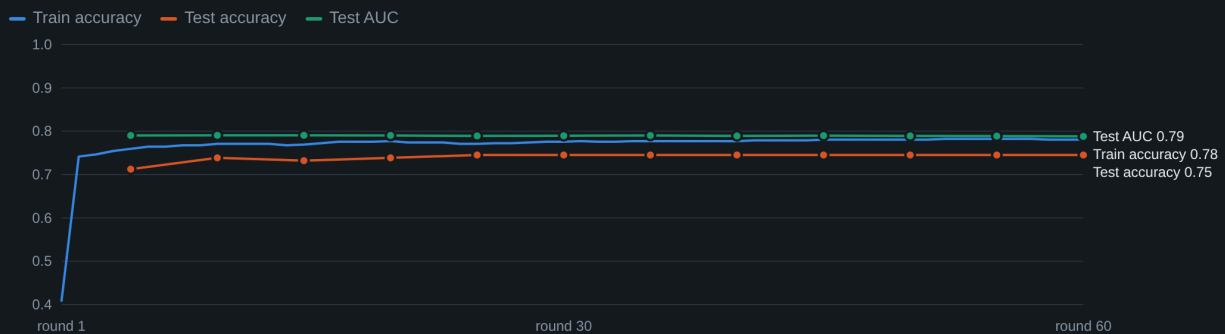
Final train loss

60

Rounds

153

Held-out people



► How to read this chart

► Table view

Reading the chart. The X axis is training rounds; the Y axis is a 0.4–1.0 score. **Blue, train accuracy:** the share of the 615 training people classified correctly at a 50% cut-off. It starts near 0.4 because an untrained model says 50% for everyone. **Orange dots, test accuracy:** the same measure on the 153 held-out people (0.75). **Green dots, test AUC:** the chance that someone who developed diabetes scores higher than someone who didn't (0.5 is chance; 0.79 is typical for this dataset). Lines flatten after about round 15, meaning the model has converged; train and test staying close means no overfitting.

Local model share

This site's slice of the federated diabetes model. The weights stay on this server.

60 training rounds · train loss 0.452 · holds the outcome label

feature	weight	
pregnancies	0.444	
blood_pressure	-0.138	
skin_thickness	0.159	
bmi	0.557	
diabetes_pedigree	0.216	
age	0.165	
bias	-0.849	

Blue bars push risk up, orange bars push it down.

From model to review

After training, each patient under review is scored the same masked way. P001 comes out at **21%**, which is the **moderate** band (cut-offs: below 15% low, below 40% moderate, otherwise high). That band becomes ledger fact **L10**, so the reasoning step can cite it next to the lab's HbA1c result.

The cut-offs are demo values and have not been clinically calibrated.

The GP's own weights, visible only on the GP's page. BMI (0.56) and pregnancies (0.44) push risk up; diastolic blood pressure pushes it slightly down.

8 Safety checks and the sabotage test

SAFEGUARD	WHAT IT STOPS
Policy before records	Out-of-purpose questions (e.g. marketing) are refused before any record is read or shown to a model.
Wire filter (allowlist)	Notes, names, dates of birth, record IDs, unknown fields and long text can never leave a department, including fields added later.
Disclosure ledger + run events	Every fact the coordinator saw is listed and numbered; nothing is hidden.
Citation check	Risk claims the ledger doesn't support are struck through. A summary with no RISKS section is blocked.
Contradiction rules	Institutions that disagree trigger escalation, which catches a compromised agent.
Re-validation on arrival	The dashboard and the SuperGrid app each re-run the wire filter on everything they receive.
Prevention-only prompt	The model recommends reviews and screenings, never diagnoses or treatment changes.
Human approval	Nothing is released without a clinician's click, and escalated reviews can't be approved.
Hard limits	At most 8 model calls and 200 training rounds per run; failures stop the run loudly instead of reporting "no findings".

The sabotage test

With sabotage on, the pharmacy agent is compromised: after it answers, its two safety answers are silently flipped to *false* (“no NSAID dispensed”, “no interaction flag”). The verifier does not trust it. The GP says the patient takes ibuprofen, but the pharmacy now claims no NSAID was dispensed, and no interaction was flagged despite active metformin. Both contradiction rules fire:

Patient P001

local deterministic rules (offline)

ESCALATE CONFLICT

engine: local deployments + offline rules

federated diabetes risk (local deployments, test AUC 0.79): **21% · moderate**

GP

6

raw records: LOCAL ONLY

PHARMACY

8

raw records: LOCAL ONLY

COMPROMISED (demo)

LAB

6

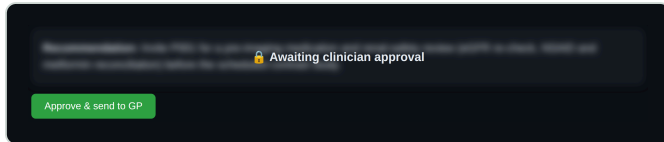
raw records: LOCAL ONLY

⚠ CONFLICT (otc_nsaid_vs_pharmacy_dispense): GP: patient self-reported OTC NSAID use, but pharmacy shows none dispensed in the last 90 days. — cites L3, L4

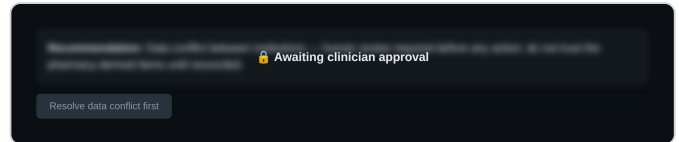
⚠ CONFLICT (metformin_nsaid_interaction_not_flagged): Pharmacy shows active metformin with no interaction flag, despite GP-reported OTC NSAID use — dispensing story is inconsistent. — cites L5, L6, L3

Both conflicts, with citations. The recommendation is replaced by “Data conflict between institutions — human review required”.

9 Clinician approval



Approved review: the recommendation stays blurred until the clinician clicks. The approval is recorded on the dashboard server with name and time.



Escalated review: the button is disabled, and the server refuses approval too (HTTP 409). The fix has to happen at the institutions.

10 Scenario: the GP advises ibuprofen

“If I see my doctor, they tell me to take ibuprofen, and I get it from the pharmacy, is information exchanged between the GP and the pharmacy?”

Inside PreventNet: no

The GP and the pharmacy never talk to each other in PreventNet. Each one separately answers the coordinator with a yes/no, and only the coordinator sees both answers, side by side in the ledger:

```
L3  gp          patient_reported_otc_nsaid = true      (medication-safety-review)
L4  pharmacy    nsaid_dispensed_90d       = true      (medication-safety-review)
```

The GP never learns what the pharmacy dispensed, the pharmacy never sees the GP's note, and neither sees the lab result. The combined picture exists only in the coordinator's ledger. It is used for one purpose (a medication-safety review) and shown to one clinician.

Outside PreventNet, a **prescription** legitimately travels from GP to pharmacy (for example an e-prescription). That is ordinary care, allowed because dispensing is its purpose. PreventNet neither replaces it nor relies on it.

One exchange does exist: during federated training, the GP sends training errors to the pharmacy. These come from the diabetes labels of the Pima cohort, not from anyone's visit or prescription. They still carry some label information; see section 13.

WHAT THE CURRENT VERSION GETS RIGHT, AND WHERE IT GOES WRONG

- **Pharmacy has a record of the ibuprofen.** Both sides say “yes”, so there is no conflict. The review flags *NSAID + reduced kidney function + contrast scan*, which is exactly the point: the GP's advice is risky given a lab result the GP may not have in view.
- **GP advised it, and the patient bought it over the counter without a patient-linked record.** The pharmacy truthfully has no record, but the current rule treats “no record” as “no”, so the verifier raises a **false conflict**.
- **Label mismatch.** The GP field is called *patient-reported*, but the rule also fires when the GP *advised* the drug. The two situations should be told apart.

Recommended handling (not yet built)

1. **Say where the GP fact came from.** Replace the yes/no with `gp_nsaid_exposure = none / advised_or_prescribed / self_reported`.
2. **Treat “no record” as “unknown”, not “no”.** If a pharmacy doesn't link over-the-counter sales to patients, it answers *unknown* for OTC and discloses nothing. No false conflict arises.
3. **Split the pharmacy fact** into prescription dispensing (a complete record) and recorded OTC sales (possibly incomplete).
4. **Narrow the conflict rule.** Raise a conflict only when two *complete* records disagree. “GP advised, pharmacy has no record” becomes a softer *adherence / OTC check*

note for the clinician, not a data conflict.

The sabotage test would keep working, because a compromised pharmacy would still contradict its own interaction-check record and the GP's prescription record. Every new field stays purpose-tagged *medication-safety-review*, so the privacy boundary does not move.

11 The four local deployments

DEPLOYMENT	ADDRESS	HOLDS	ITS OWN PAGE SHOWS
GP	127.0.0.1:8101	Records + 6 Pima columns + the diabetes label	Raw records, disclosure log, its model weights
PHARMACY	127.0.0.1:8102	Dispensing records + synthetic column	Same, plus the sabotage switch
LAB	127.0.0.1:8103	Lab results + glucose, insulin	Same
COORDINATOR	127.0.0.1:8100	Nothing raw	The dashboard: data explainers, federated training, reviews, the live SuperGrid log and Endeavor's answer, approvals

Department deployments

Each department runs its own server, holds its own data, and has its own overview page. Open one to see what stays local.

GP

● online

<http://127.0.0.1:8101> · raw data stays here

PHARMACY

● online

<http://127.0.0.1:8102> · raw data stays here

LAB

● online

<http://127.0.0.1:8103> · raw data stays here

Cohort alignment: all three departments report the same patient-ID fingerprint ✓

Dashboard header. The three departments are online, and their patient-ID fingerprints match, so the cohorts line up.

PHARMACY · LOCAL DEPLOYMENT

Local deployment · raw data never leaves this server

This page is served by the site itself. The coordinator can only call /ask (categorical answers) and /f1/* (masked scores); it never loads this view.

Demo: compromise this agent

When on, this site's answers about NSAID dispensing and interaction flags are forced to *false* after the rules run. The coordinator's verifier should catch the contradiction.

Turn sabotage ON

What this site disclosed

Local audit log of every answer and refusal sent to the coordinator.

time	patient	event
12:46:53	P001	disclosed interacting_dispense_flag = true (medication-safety-review)
12:46:53	P001	disclosed metformin_active = true (renal-risk-review)
12:46:53	P001	disclosed nsaid_dispensed_90d = true (medication-safety-review)

Local model share

This site's slice of the federated diabetes model. The weights stay on this server.

60 training rounds

feature	weight
antihypertensive_dispensed	0.062

Blue bars push risk up, orange bars push it down.

Patient records (raw, local only)

patient	kind	date	details
P001	dispense	2026-01-10	drug: metformin · dose_mg: 500
P001	dispense	2026-02-10	drug: metformin · dose_mg: 500
P001	dispense	2026-03-10	drug: metformin · dose_mg: 500
P001	dispense	2026-04-10	drug: metformin · dose_mg: 500

Pharmacy's own page, served only by the pharmacy: the sabotage switch, every answer it disclosed, its single model weight, and its raw dispensing records. The coordinator never loads this page.

Review engines on the dashboard

- **SuperGrid AgentApp** (default): starts the full federated app in Flower's cloud, streams its log, then shows what Flower Endeavor answered, the verifier's explanation, the run ID and the training chart from inside the run. Takes about 1–2 minutes.
- **SuperGrid Endeavor over local facts**: the local departments answer; only their short facts go to the cloud for reasoning. No record leaves the laptop.
- **Offline**: everything local, with a fixed template instead of a model. Instant.

12 Running it

```
# one-time setup
uv sync
uv run pytest -q                # 37 offline tests
uv run flwr build
uv run flwr login supergrid

# the product: one federated review on SuperGrid
uv run flwr run . supergrid --stream | tee run_happy.log
uv run flwr run . supergrid --run-config 'demo.sabotage=true' --stream | tee run_sabotage.log
python scripts/extract_result.py run_happy.log      # → viewer/latest.json

# the window: four local deployments
uv run python -m local.launch    # then open http://127.0.0.1:8100
```

SETTING	DEFAULT	MEANING
patient.id	P001	Patient to review (P001–P003)
demo.sabotage	false	Compromise the pharmacy agent
fl.rounds / fl.lr	60 / 0.5	Federated training rounds (1–200) and learning rate
model.ref	flower-endeavor-v1.0	Model; falls back to openai/gpt-5.6-sol if not found
review.ledger	""	Set by the dashboard to reason over local facts only

13 Limits and open points

- **One process on SuperGrid.** Inside the SuperGrid run, the departments are separated by code structure, and all three data folders ship in one app bundle. Physically separate departments would need one Flower node per department.
- **Plain vertical FL.** The training errors the GP sends carry some label information, and the coordinator sees each person's summed score. Production systems add encryption or noise.
- **Synthetic data.** Review records are synthetic, and the pharmacy's Pima column is made up. Risk-band cut-offs are uncalibrated.
- **OTC purchases.** These can raise a false conflict today; section 10 describes the fix.
- **Model ID source.** The Endeavor ID was taken from a public Flower example app rather than the official documentation.
- **Platform quirks found on real runs.** Every run event needs a `type` field.
`context.locked()` does not exist in Flower 1.37. App bundles can't contain `.csv` files.

PreventNet · nishankks/preventnet 0.1.0 · Flower 1.37 · Screenshots from the local deployments (offline engine), 16 September 2026 · MIT licence